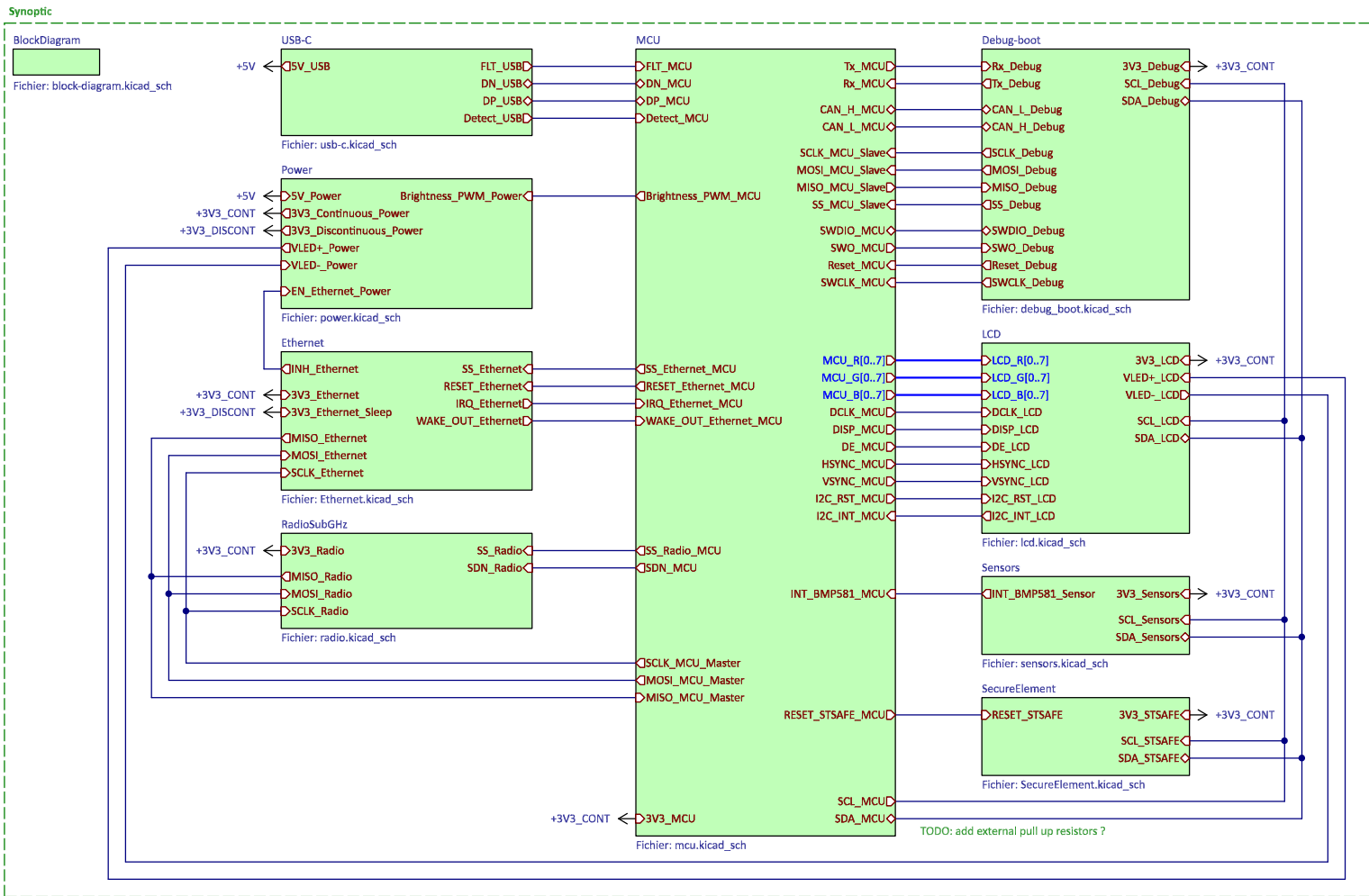


Weather station POC



Revision			
Rev.	Description	Date	Approved
A	Project creation	01/11/2025	JCR

Julien Cruvieux

Sheet: /

File: weather-station-poc.kicad_sch

Title: Weather station POC

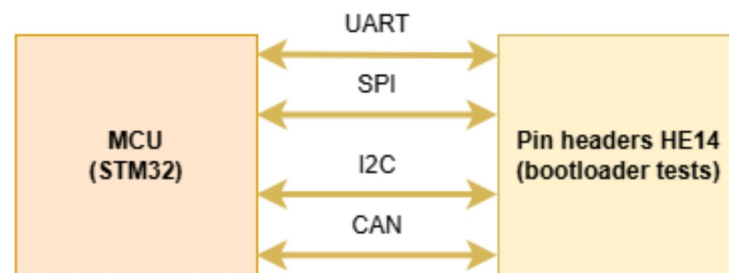
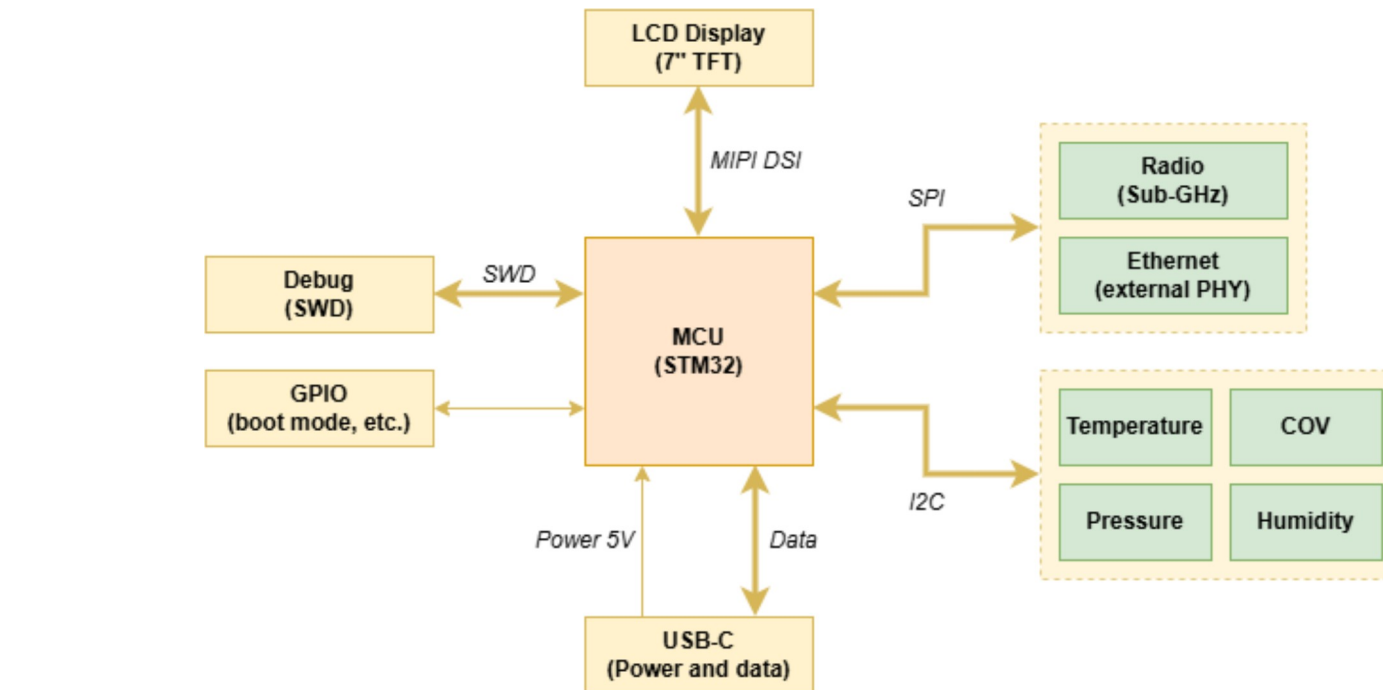
Size: A4

Date: 2025-10-26

KiCad E.D.A. 9.0.6

Rev: A

Id: 1/11



Sheet: /BlockDiagram/
File: block-diagram.kicad_sch

Title:

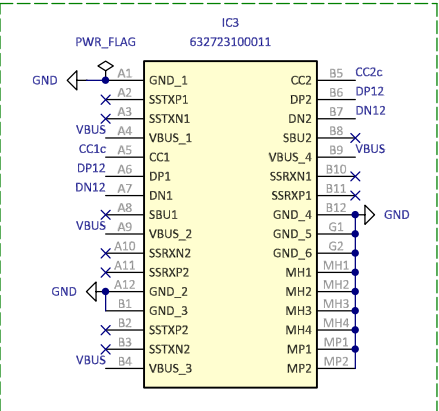
Size: A4
KiCad E.D.A. 9.0.6

Date:

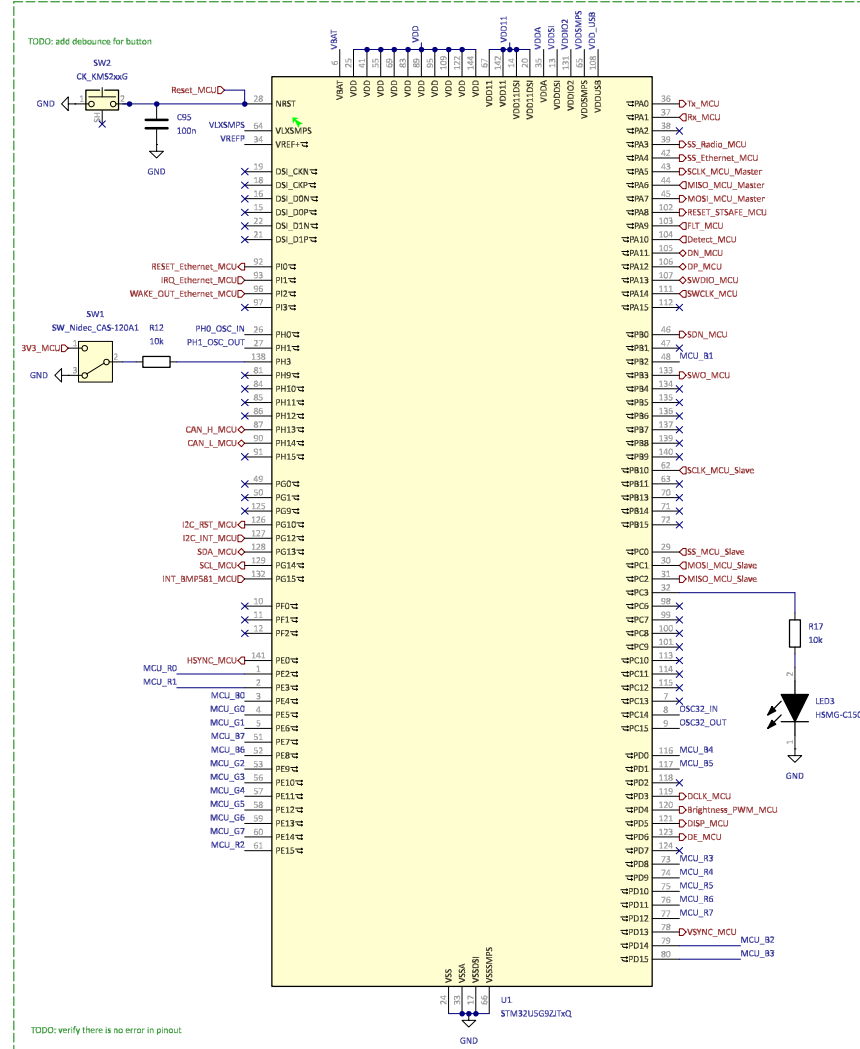
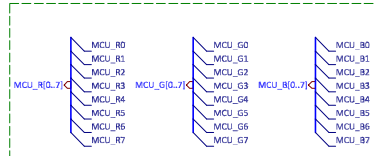
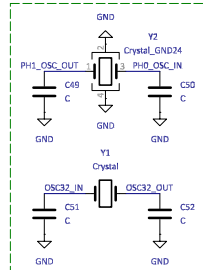
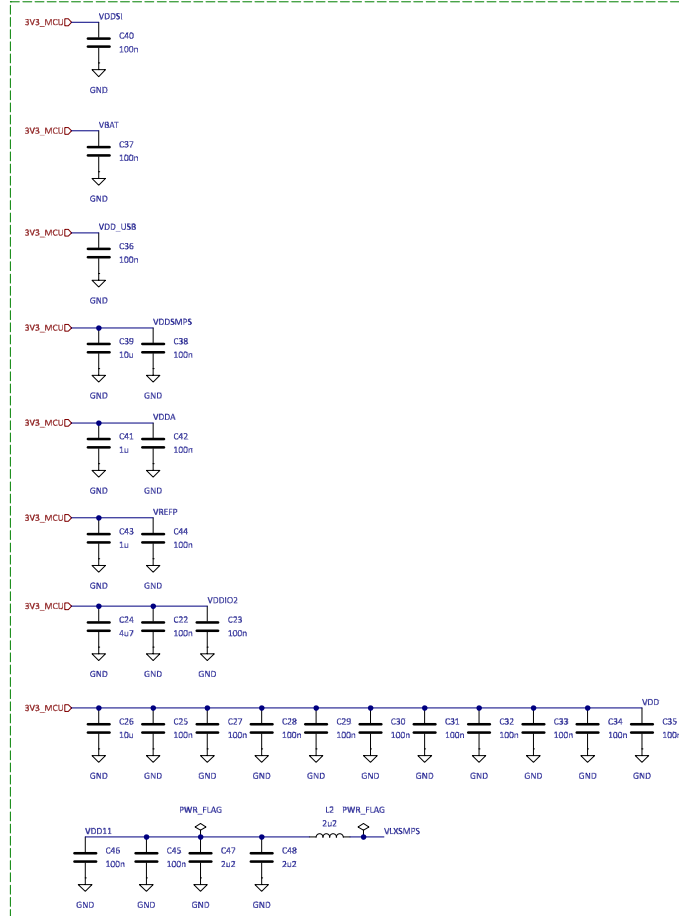
Rev:
Id: 2/11

USB-C

USB-C connector right angle



MCU



Sheet: /MCU/
File: mcu.kicad_sch

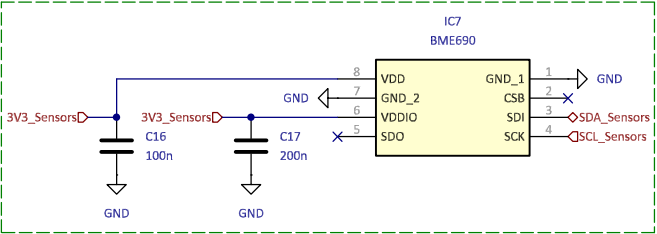
Title:

Size: A3
KiCad E.D.A. 9.0.6

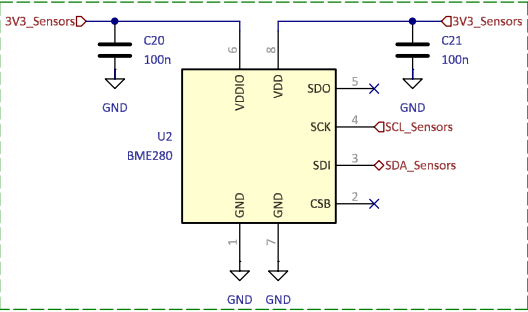
Rev:
Id: 4/11

Sensors

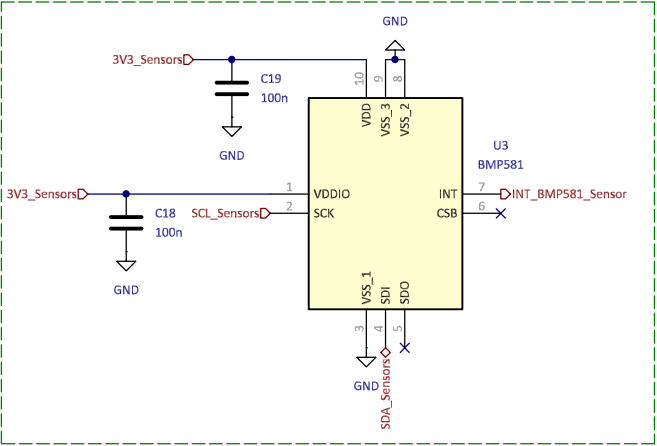
Quality air sensor



Humidity sensor



Pressure and temperature sensor



Sheet: /Sensors/
File: sensors.kicad_sch

Title:

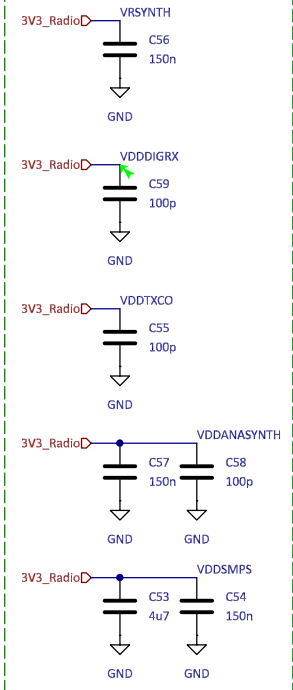
Size: A4
KiCad E.D.A. 9.0.6

Date: 2025-11-01

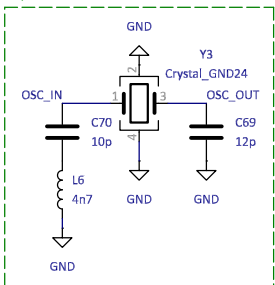
Rev:
Id: 5/11

Sub-GHz radio

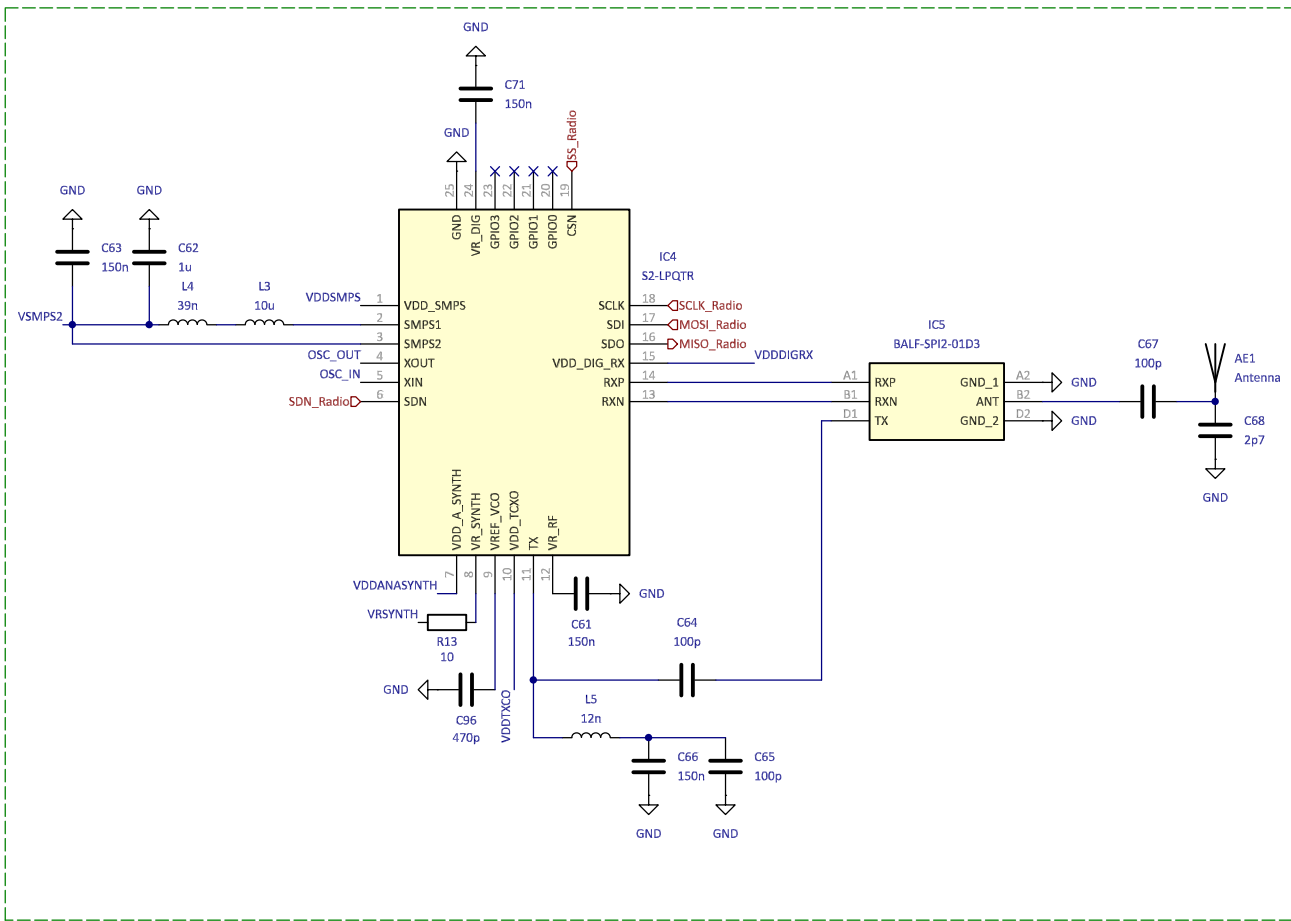
Decoupling capacitors



Crystal oscillator



Sub-GHz module



Sheet: /RadioSubGHz/
File: radio.kicad_sch

Title:

Size: A4

Date: 2025-11-01

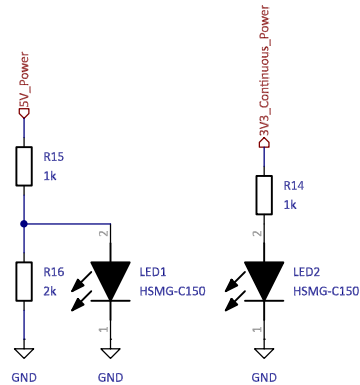
KiCad E.D.A. 9.0.6

Rev:

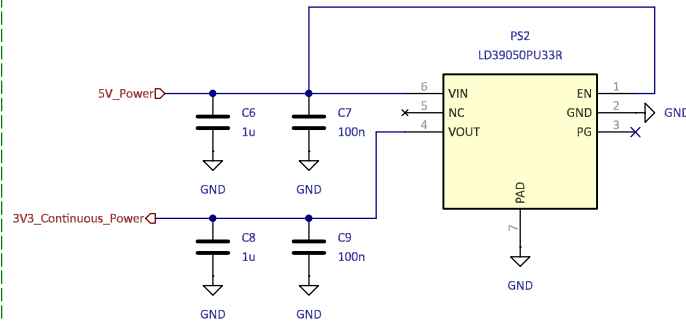
Id: 6/11

Power supplies

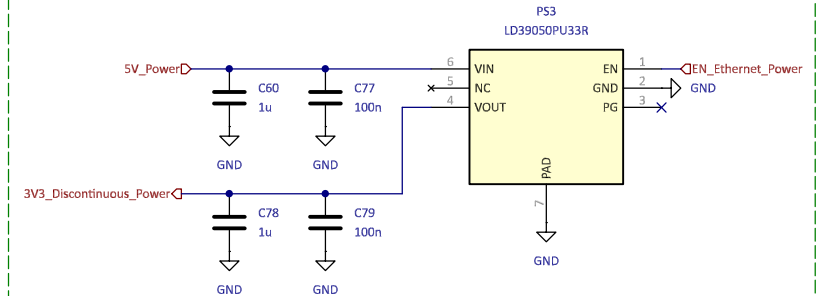
Power LED status



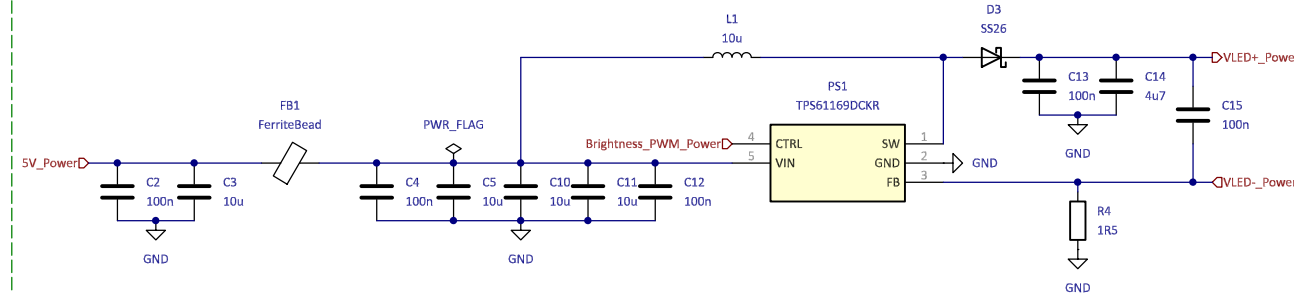
LDO 5V -> 3.3V



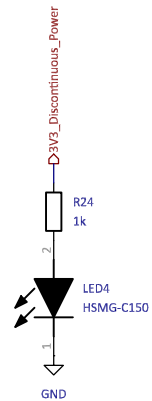
LDO 5V -> 3.3V



LED driver



The TP561169 WLED (White LED) driver is a boost converter regulated in current through FB pin. The driver uses a 204mV voltage reference and the resistor on FB pin set the current threshold -> $204\text{mV} / 1.5\text{ ohm} = 136\text{ mA}$. To have full brightness on 5" Riverdi screen, 140 mA are needed. To control the brightness, a PWM signal is injected on CTRL pin. 100% duty cycle is full brightness, 50% half brightness and so on ...



Sheet: /Power/
File: power.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 7/11

Debug

SWD connector (MIPI 10)

Conn_ARM_JTAG_SWD_10

TODO: see what resistor set on SWD

Communication interface connectors

TODO: Sub d 9pts connector is very big on a PCB, maybe replace by a pin header

Sheet: /Debug-boot/ File: debug_boot.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 8/11

Title:

Size: A4

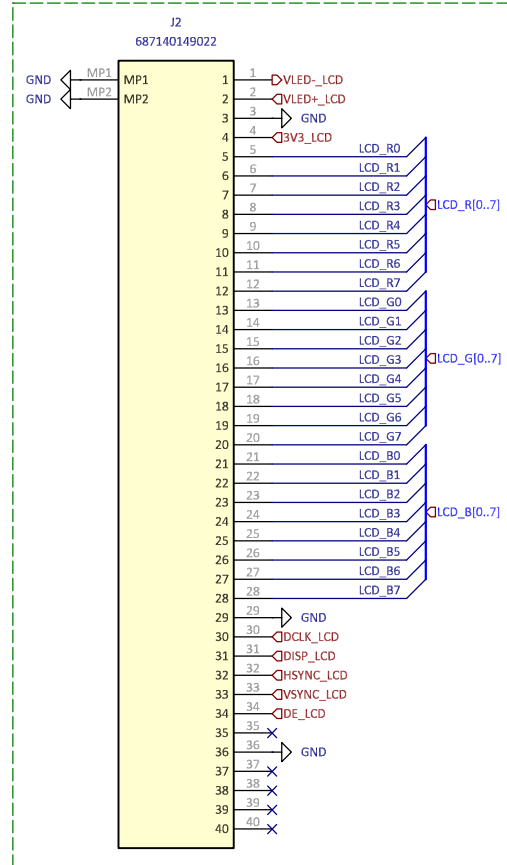
KiCad E

Id: 8/11

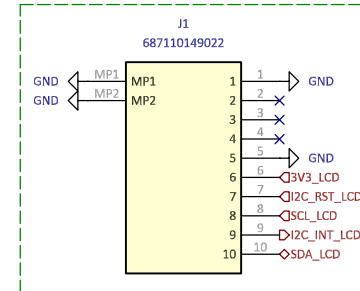
4

TFT LCD screen

TFT LCD 40 pins driver connector



TFT LCD 10 pins touch connector



TODO: think about ferrite beam on LCD connectors

Sheet: /LCD/
File: lcd.kicad_sch

Title:

Size: A4

Date:

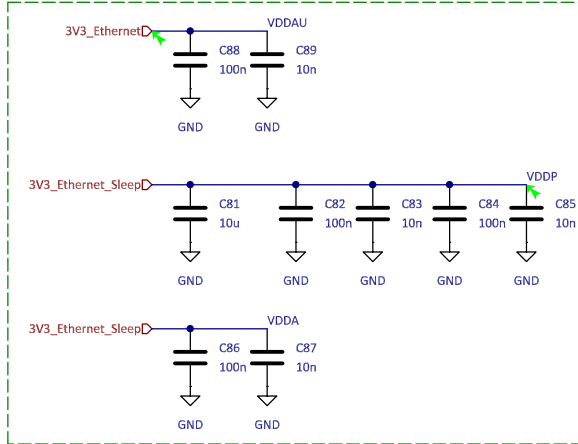
KiCad E.D.A. 9.0.6

Rev:

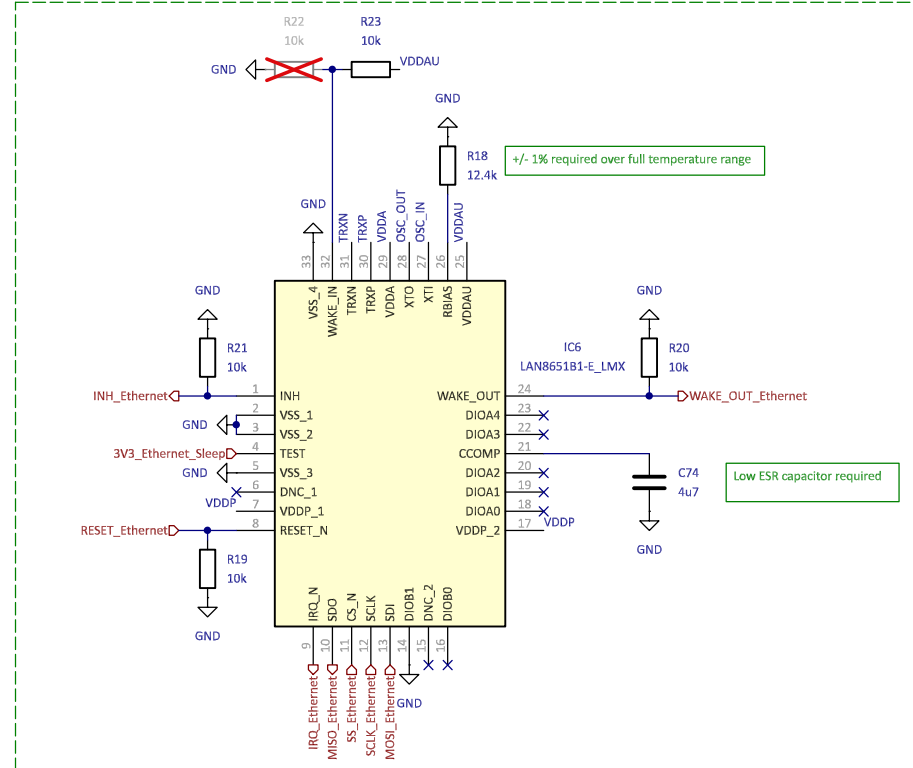
Id: 9/11

Ethernet PHY

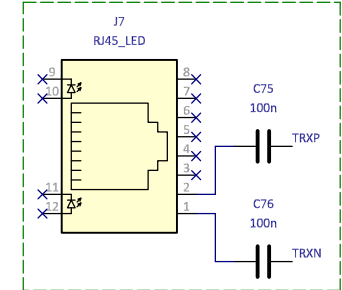
Ethernet LAN8651 decoupling capacitors



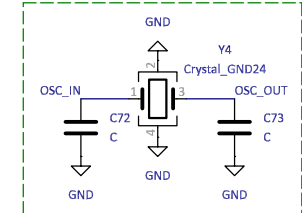
External Ethernet PHY



RJ45 Connector for LAN8651



LAN8651 25MHz oscillator



Sheet: /Ethernet/
File: Ethernet.kicad_sch

Title:

Size: A4

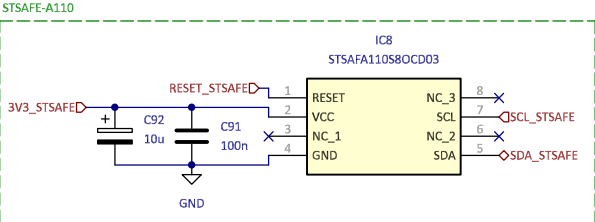
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 10/11

Secure element (STSAFE)



Sheet: /SecureElement/
File: SecureElement.kicad_sch

Title:

Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.6

Id: 11/11